Mathieu Feraud

Introduction

Echo chambers have emerged as a defining feature of contemporary online discourse, particularly on social media platforms. In political and social communication networks, echo chambers are understood as environments in which users predominantly interact with like-minded peers, reinforcing shared beliefs while minimizing exposure to dissenting perspectives. Such clustering can lead to increased polarization, reinforced confirmation bias, and the rapid spread of narrow narratives among participants who primarily consume information that echoes their pre-existing views.[1,2]

Research across multiple platforms—including Twitter, Facebook, Reddit, and Gab—has documented how interaction patterns and algorithmic content curation contribute to the formation of ideologically homogeneous communities. Studies tracking millions of pieces of content related to controversial topics show that homophily in interaction networks and preferential diffusion toward like-minded users are central drivers of echo chamber effects in online environments[2]. Theoretical and empirical work suggests that both user behaviour (e.g., selective exposure and unfollowing) and platform dynamics (e.g., personalized feeds and recommendation systems) can accelerate the emergence of segregated clusters of discourse, even when minimal influence and unfriending behaviour are present[3].

The prevalence of echo chambers has important implications for public discourse and democratic deliberation. Social media-based echo chambers can reinforce belief systems, limit exposure to counterarguments, and facilitate the spread of misinformation or contested claims, particularly on highly politicized topics. Empirical studies of COVID-19 communication, for example, find that political echo chambers were especially dense in certain communities, reinforcing exposure to information consistent with users' existing views and contributing to asymmetric polarization[4].

Crucially, echo chambers do not manifest uniformly across contexts, topics, or platforms. Conceptual and empirical work shows that echo chambers can vary in intensity, structure, and mechanism, reflecting differences in homophilic clustering, selective exposure dynamics, and network diffusion patterns. The concept of echo chambers itself has been operationalized in multiple ways, and research highlights the complexity of underlying mechanisms such as selective exposure, content distribution bias, and confirmation bias[1].

Given this diversity of forms and mechanisms, a binary measure of "echo chamber present/absent" is inadequate. To understand how and why certain communities exhibit echo chamber dynamics, it is critical to examine not only whether echo chambers exist, but also what qualities define their formation and maintenance. This motivates the development of ChamberCheck, a multidimensional analytical framework designed to assess not just the presence of echo chamber effects in a subreddit but the specific social dynamic features that contribute to their emergence and persistence.

Methodology

Data Source

This study analyses online discussion communities hosted on Reddit and Facebook. While these platforms differ in interface and moderation mechanisms, subreddits and Facebook Groups share key structural similarities: topic-focused communities, user-generated posts, threaded comments, and engagement signals (e.g., reactions, replies).

To enable cross-platform analysis under a unified framework, data from both platforms are extracted using their respective APIs and stored in a common schema. Hereafter, subreddits and Facebook Groups are jointly referred to as groups. Data Extraction

To be able to analyse and compare subreddit or Facebook groups, The following data will be extracted from these:

2. Data Extraction

For each group, the following data are collected.

2.1 Raw Data

Group-level metadata

- Group name
- Subscriber or member count
- Description and rules
- Creation date

Post-level data

- Parent group
- Title
- Timestamp
- Post content (text or media reference)
- Engagement signals (e.g., upvotes, reactions)
- Associated comments

Comment-level data

- Parent post
- Parent comment (if applicable)

- Timestamp
- Comment text

Content Normalisation and Transformation

Because posts may contain non-textual media, all content is transformed into a textual representation prior to analysis to ensure compatibility with large language models (LLMs).

The following transformations are applied:

- **Images**: Images are described using an LLM-generated caption, capped at 300 words per image.
- **YouTube videos**: Video transcripts are extracted where available and summarised to a maximum of 1,000 words.
- **External links**: Linked webpages are scraped for textual content and summarised to a maximum of 1,000 words.
- **Polls**: Poll questions and aggregate results are extracted verbatim.

For topic-sensitive posts (e.g., politics, war, identity, or country-related discussions), the LLM additionally provides:

- A brief topic description
- A coarse topic valence score on a continuous scale from −10 (extremely negative) to +10 (extremely positive)

This auxiliary signal is used only for topic characterisation, not for stance inference.

LLM-Based Foundational Attribute Extraction

Each post and comment is independently analysed using an LLM to extract foundational discourse attributes. The LLM prompt for comments can be seen at S1 Comment LLM Prompt. These attributes form the basis for all derived echo chamber metrics.

The LLM is constrained to:

- Use only information explicitly present in the text
- Avoid external knowledge or factual verification
- Return "N/A" when an attribute cannot be inferred

The extracted attributes include:

- Topic label and confidence
- Topic stance (continuous $\in [-1, 1]$) and confidence

- Substantiveness
- Discrediting or dismissive tone
- Topic Sensitivity
- Defensive framing
- Toxicity
- Epistemic risk
- Politeness
- Emotion intensities (anger, fear, outrage, anxiety, disgust)
- Misinformation likelihood (as rhetorical risk, not factual correctness)

All scalar attributes are continuous $\in [0, 1]$ unless otherwise stated.

Derived Echo Chamber Metrics

Echo chambers are treated as multidimensional phenomena. ChamberCheck measures nine metrics that operationalise five behavioural antecedents of echo chamber formation identified in the systematic review literature. All reply-level metrics use Option A aggregation: replies to the same parent comment are averaged at parent level before cross-group comparison, so each parent comment contributes one observation regardless of how many replies it received.

Shared design decisions applied consistently across all metrics: (1) neutral stance threshold — comments with $|\text{stance}| \leq 2$ are excluded from all stance-based classifications; (2) majority definition — the majority group must hold ≥ 60% of classified comments in a thread or topic; (3) weighting variable α_t = count of base observations in topic t, used consistently for all topic-to-subreddit aggregations.

Avoidance & Discreditation: CSS (Counter-Stance Silence Rate) measures passive avoidance — whether counter-stance comments are systematically left without any reply. CSEQ (Counter-Stance Engagement Quality) addresses the active discreditation mechanism: when counter-stance comments do receive replies, how substantive is that engagement? Together they characterise the full avoidance/discreditation spectrum.

Spiral of Silence: SBI (Stance Balance Index) measures whether both sides of an argument are represented in the comment pool, reported per topic on a scale of 0 (completely one-sided) to 0.5 (perfectly balanced). MSDG (Minority Stance Defensiveness Gap) tests the core spiral of silence prediction — whether minority-stance commenters use more defensive language than majority-stance commenters.

Selective Exposure: RDB (Reply Direction Bias) measures whether users preferentially reply to same-stance comments relative to what would be expected from the thread's stance composition. uRDB (User-Level Reply Direction Bias) applies the same logic at the individual-user level, preventing high-volume users from dominating the aggregate and revealing whether individual users engage across the aisle.

Emotions: EAS (Emotional Amplification Score) measures whether the community's upvote mechanism rewards emotional content, computed as Spearman ρ between upvote counts and each emotion score (anger, anxiety, disgust). CSAD (Cross-Stance Anger Differential) measures whether users are angrier when replying to opponents than to allies, reported directionally for Majority→Minority and Minority→Majority interactions alongside a same-stance baseline.

Affective Polarization: TD (Toxicity Differential) directly operationalises affective polarization — hostile language directed specifically at out-group members. It follows the same directional structure as CSAD but uses the toxicity score rather than anger. CSAD and TD share the same aggregation pass; divergence between them (high CSAD, low TD) suggests emotional engagement without explicitly hostile language.

1a. CSS — Counter-Stance Silence Rate

Antecedent: Avoidance & Discreditation | LLM variables: stance (all levels), reply_count (structural)

For each thread, comments are classified as counter-stance or same-stance relative to the thread majority. The silence (zero-reply) rate is computed for each group. The gap is the avoidance signal. Requires ≥ 5 counter-stance comments per thread.

counter = comments whose stance sign differs from thread majority

same = comments whose stance sign matches thread majority

CSS_thread = silence_rate(counter) - silence_rate(same)

CSS_topic = weighted mean across threads (α = counter-stance comment count)

CSS_subreddit = weighted mean across topics

1b. CSEQ — Counter-Stance Engagement Quality

Antecedent: Avoidance & Discreditation | LLM variables: discrediting, evidence_quality, reasoning_depth, stance (all levels)

For each counter-stance parent comment, the mean discrediting, evidence quality, and reasoning depth are computed across all replies. The key diagnostic value is the Spearman ρ between discrediting and evidence_quality: a strong negative correlation means the community dismisses opposing views without substantive engagement. Reported separately for Majority→Minority and Minority→Majority directions.

For each counter-stance parent p:

avg_discrediting(p) = mean(discrediting of all replies to p)

avg_evidence_quality(p) = mean(evidence_quality of all replies to p) avg_reasoning_depth(p) = mean(reasoning_depth of all replies to p)

CSEQ_subreddit = weighted mean across topics (α = counter-stance parent count)

2a. SBI — Stance Balance Index

Antecedent: Spiral of Silence | LLM variables: stance (top-level comments sufficient)

For each topic, counts positive-stance and negative-stance comments (excluding near-neutral). The minority proportion is the smaller count divided by the total. Reported per topic only — do not aggregate to subreddit level. No confidence intervals; descriptive only.

pos = count of comments with stance > +2

neg = count of comments with stance < -2

SBI_t = min(pos, neg) / (pos + neg) range: 0.0 to 0.5

2b. MSDG — Minority Stance Defensiveness Gap

Antecedent: Spiral of Silence | LLM variables: stance (all levels), defensiveness

For each topic, identifies the minority stance group by count and computes whether that group uses more defensive language than the majority group. Excludes topics with SBI > 0.4 (unstable minority definition) and topics with fewer than 10 minority comments.

minority_group = whichever of pos_group / neg_group is smaller

MSDG_t = mean(defensiveness | minority) - mean(defensiveness | majority)

MSDG_subreddit = weighted mean across topics (α = minority comment count)

3a. RDB — Reply Direction Bias

Antecedent: Selective Exposure | LLM variables: stance (all levels), reply structure (structural)

For each parent comment, computes the fraction of replies directed at same-stance commenters. Reported separately for majority-stance users (RDB_pro) and minority-stance users (RDB_con). A value of 0.5 indicates random stance-blind reply behaviour; above 0.5 indicates preferential same-stance engagement.

rate_same(p) = same-stance replies / all replies to p

RDB_thread = mean(rate_same) across all parents in thread

RDB_subreddit = weighted mean across topics (α = parent comment count)

3b. uRDB — User-Level Reply Direction Bias

Antecedent: Selective Exposure | LLM variables: stance (all levels), reply structure (structural)

For each user with at least 4 replies in a thread, computes the proportion of their replies directed at same-stance parents. These per-user values are averaged within the thread before aggregation, preventing high-volume users from dominating.

For each user u with >= 4 replies in thread t:

uRDB_user(u,t) = count(replies to same-stance parents) / count(all classified replies)

uRDB_thread = mean(uRDB_user) across eligible users in t

uRDB_subreddit = weighted mean across topics (α = eligible user count)

4a. EAS — Emotional Amplification Score

Antecedent: Emotions | LLM variables: anger, anxiety, disgust (LLM JSON), upvotes (structural)

Computes the Spearman rank correlation between upvote counts and each emotion score across all comments in the subreddit. A positive ρ means the community's upvote mechanism rewards emotional content. Computed at subreddit level as the primary unit; topic-level EAS is a secondary analysis only (minimum 50 comments per topic). FDR correction (Benjamini-Hochberg) applied globally.

For each emotion e in {anger, anxiety, disgust}:

EAS_e = spearmanr(upvotes, e) across all comments in subreddit

Bootstrap CIs: resample at comment level, 1000 iterations

4b. CSAD — Cross-Stance Anger Differential

Antecedent: Emotions | LLM variables: anger (LLM JSON), stance (all levels)

For each parent comment, anger is averaged across all replies and classified by direction (Majority→Minority, Minority→Majority, or same-stance). The three-way comparison reveals whether cross-stance interactions are angrier than same-stance baseline interactions, and whether one direction is more hostile than the other.

avg_anger(p) = mean(anger across all replies to p)

CSAD_aggregate = mean(avg_anger | cross-stance) - mean(avg_anger | same-stance)

CSAD_Maj->Min and CSAD_Min->Maj reported separately

CSAD_subreddit = weighted mean across topics (α = parent comment count)

5. TD — Toxicity Differential

Antecedent: Affective Polarization | LLM variables: toxicity (LLM JSON), stance (all levels)

Identical structure to CSAD but uses toxicity score instead of anger. Toxicity captures hostile, aggressive, or abusive language directed at the parent comment or author (insults, slurs, dehumanisation, threats). CSAD and TD share the same aggregation pass.

avg_toxicity(p) = mean(toxicity across all replies to p)

TD_aggregate = mean(avg_toxicity | cross-stance) - mean(avg_toxicity | same-stance)

TD_Maj->Min and TD_Min->Maj reported separately

TD_subreddit = weighted mean across topics (α = parent comment count)

Prompt validation and annotation agreement

To assess the reliability of the LLM-based annotation pipeline, a manual validation study will be conducted. A random sample of 50 posts or comment threads will be independently annotated by the author using the same attribute definitions provided to the LLM. LLM outputs will then be compared against manual annotations using appropriate agreement measures, such as correlation or mean absolute error for continuous attributes, and agreement rates or chance-corrected statistics where applicable.

This validation is intended to evaluate construct alignment and prompt consistency, rather than to establish ground-truth correctness. The sample size is chosen to balance feasibility with coverage across multiple topics and discourse styles.

Discrepancies observed during this process will be used to refine prompt wording and definitions prior to full-scale analysis.

1. Hartmann, D., Wang, S. M., Pohlmann, L. & Berendt, B. A systematic review of echo chamber research: comparative analysis of conceptualizations, operationalizations, and varying outcomes. *J. Comput. Soc. Sci.* **8**, 52 (2025).

2. Cinelli, M., De Francisci Morales, G., Galeazzi, A., Quattrociocchi, W. & Starnini, M. The echo chamber effect on social media. *Proc. Natl. Acad. Sci. U. S. A.* **118**, e2023301118 (2021).

3. Sasahara, K. *et al.* Social influence and unfollowing accelerate the emergence of echo chambers. *J. Comput. Soc. Sci.* **4**, 381–402 (2021).

4. Jiang, J., Ren, X. & Ferrara, E. Social Media Polarization and Echo Chambers in the Context of COVID-19: Case Study. *JMIRx Med* **2**, e29570 (2021).

5. Neely, S. R. Politically Motivated Avoidance in Social Networks: A Study of Facebook and the 2020 Presidential Election. *Soc. Media Soc.* **7**, 20563051211055438 (2021).

Data Pipeline

The pipeline consists of eight sequential stages. Each stage reads the output of the previous one. All hardcoded thresholds and prompts are centralised in src/ChamberCheck/constants.py.

Stage 1 — Scrape Posts

Post metadata (title, author, upvotes, comment count, URL, creation date) is collected for each subreddit using Reddit's public JSON API with no authentication required. Posts are sorted by the configured method (default: all-time top) and filtered to a minimum comment count before being written to disk. A new output directory is auto-created for each run.

Output: data/raw/scrape_NNN/posts.json

Stage 2 — Analyse Post Titles

Each post title is sent to an LLM with a classification prompt (PROMPT_POST_TITLE_ANALYSIS) that assigns a topic from a fixed three-level taxonomy (top → mid → leaf) and a discussion score between 0 and 1. The discussion score reflects the expected degree of substantive opinion-driven disagreement the post is likely to generate. Only post titles are used (not bodies), to avoid leaking subreddit-identifying context to the LLM.

Output: data/raw/scrape_NNN/posts_analysis/analysis_NNN.json

Stage 3 — Preprocess Posts (Filter & Rank)

Raw post data is joined with the LLM topic analysis and filtered in sequence: minimum comment count, non-UNCLEAR topic, and minimum discussion score. A topic-group peer filter then requires that each topic group contain at least a minimum number of posts to ensure sufficient within-topic comparisons. Surviving posts are ranked by discussion score and capped at top_n_per_subreddit per community.

Output: data/raw/scrape_NNN/pre_process/pre_process_NNN.json

Stage 4 — Scrape Comments

The full comment tree for each selected post is fetched via Reddit's public JSON API, with pagination and rate limiting. For each comment the scraper extracts: comment_id, parent_id, body, author, upvotes, created_at, and depth. Comments are aggregated into a single file grouped by post.

Output: data/raw/scrape_NNN/comments_NNN.json

Stage 5 — Preprocess Comments

Comments shorter than 20 characters are removed along with their entire subtrees. The remaining comments are sampled using trunk-based random sampling: whole conversation threads are added until a cap of 100 comments per post is reached or all threads are exhausted. This preserves conversation structure while keeping LLM annotation costs bounded.

Output: data/raw/scrape_NNN/comments_filtered_NNN.json

Stage 6 — Analyse Comments (LLM Annotation)

Each filtered comment is sent to an LLM (claude-haiku-4-5-20251001) along with the full prior thread context. The prompt extracts structured discourse attributes: comment type, topic label, stance value and rationale, epistemic risk sub-scores (claim strength, evidence quality, reasoning depth), toxicity, discrediting, defensiveness, civility, and emotion intensities (anger, anxiety, disgust). All scores are on a 0–10 scale. The static instruction block is cached as the Anthropic system prompt for approximately 90% cost reduction across repeated calls. Results are written incrementally via atomic .tmp → rename to survive partial runs.

Output: data/output/scrape_NNN/comment_analysis_NNN.json

Stage 7 — Compute V3 Metrics

The nine V3 echo-chamber metrics are computed from the LLM-annotated comment data by V3Metrics.from_files(). This joins the comment analysis output with the filtered comments file to enrich LLM scores with scraped metadata (community, parent_id, upvotes, author, depth). Stance polarity is determined from the topic.stance.value field; comments with |stance| ≤ 2 are treated as neutral and excluded from all stance-dependent metrics. The majority stance is whichever side holds ≥ 60% of classified comments per topic. Benjamini–Hochberg FDR correction is applied globally across all p-values.

Output: data/output/scrape_NNN/v3_metrics_NNN.json

Stage 8 — Visualisations

The plots are generated by scripts that read the per-subreddit metric files and produce publication-ready figures using matplotlib. Two sets of plots are produced per run: comment-level descriptive plots (distributions of toxicity, discrediting, defensiveness, civility, stance, emotion, comment type, and epistemic risk) and V3 echo-chamber metric plots (CSS, CSEQ, SBI, MSDG, RDB, uRDB, EAS, CSAD, TD). When combining data from multiple scraping runs, subreddits are placed on the same axes with a vertical divider separating the groups.

Output: data/output/scrape_NNN/plots/ and data/output/scrape_NNN/v3_plots/

Results — Visualisations

The following plots show results from 26,996 LLM-annotated comments across 9 subreddits. All metrics are computed independently per subreddit; no cross-subreddit pooling is performed. Error bars show 95% confidence intervals throughout. See

Supplementary Material S5 for details on the two data collection runs and how they were combined.

1a. CSS — Counter-Stance Silence Rate

CSS measures whether minority-stance comments receive replies at a lower rate than majority-stance comments within the same topic. It is computed as the silence rate of minority-stance comments (fraction with zero replies) minus the silence rate of majority-stance comments. A positive value would indicate that minority views are being ignored more frequently — a signal of avoidance-based echo-chamber dynamics. A negative value means minority comments are actually replied to at a higher rate than majority comments.

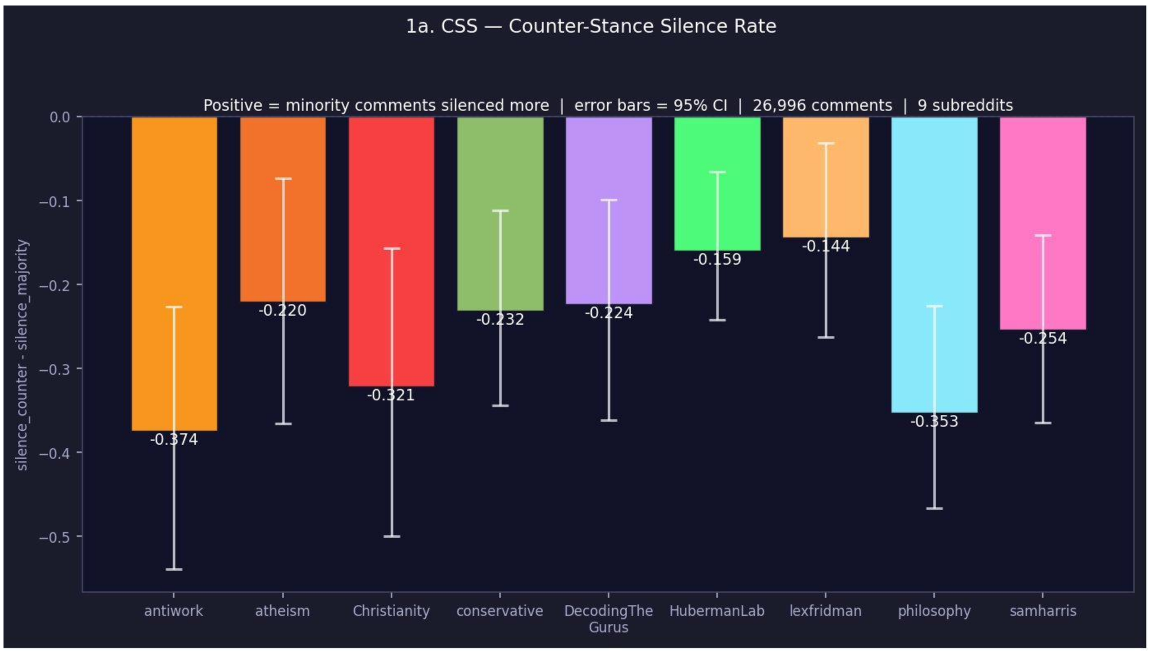

Figure 1a. Counter-Stance Silence Rate (CSS) per subreddit (n = 26,996 comments, 9 subreddits). Y-axis: silence rate of minority comments minus silence rate of majority comments. All communities show negative CSS, meaning minority-stance comments attract more replies than majority-stance comments, not fewer. Error bars show 95% CI.

1b. CSEQ — Cross-Stance Engagement Quality

CSEQ characterises the quality of replies directed at cross-stance comments. For each subreddit, three sub-metrics are computed for replies going from majority-stance to minority-stance commenters: mean discrediting score, mean evidence quality, and mean reasoning depth. A community with high discrediting but low evidence quality in cross-stance replies is engaging with opposing views dismissively rather than substantively. The same three sub-metrics are also computed for the minority→majority direction (shown as lighter bars of the same colour).

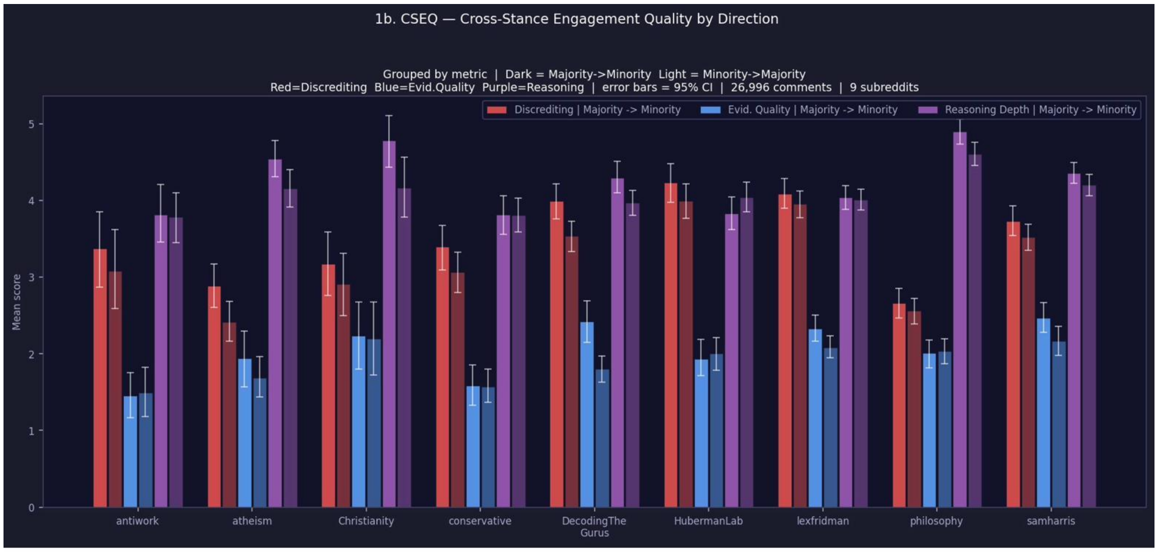

Figure 1b. Cross-Stance Engagement Quality (CSEQ) per subreddit. Bars show mean score per metric and direction: dark bars = Majority→Minority, light bars = Minority→Majority. Red = Discrediting, Blue = Evidence Quality, Purple = Reasoning Depth. Error bars show 95% CI.

2a. SBI — Stance Balance Index

SBI measures how evenly balanced each discussion topic is between positive and negative stances, computed across all subreddits. It ranges from 0 (completely one-sided, all comments share the same stance) to 0.5 (perfectly balanced). Topics with fewer than 10 classified comments are excluded. Each bar shows the topic label, its SBI value, the total number of classified comments (n=), and the positive/negative stance split in parentheses.

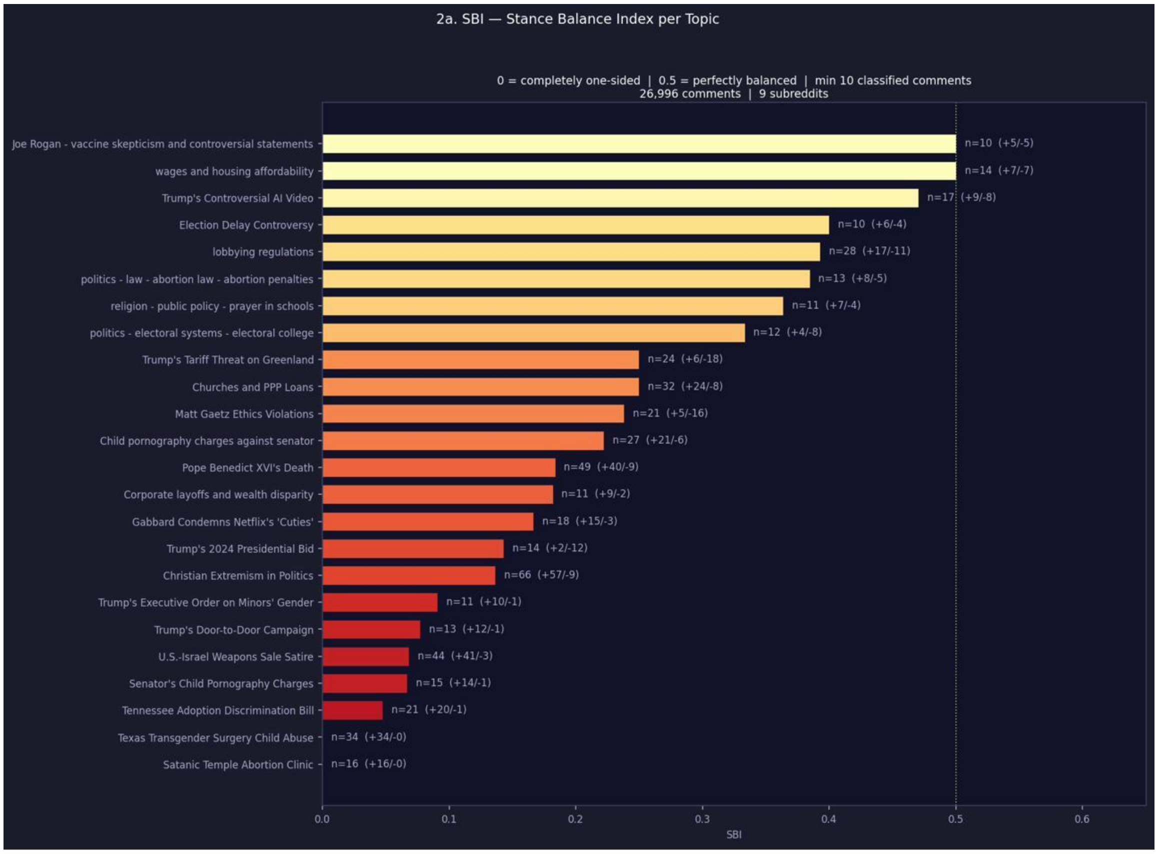

Figure 2a. Stance Balance Index (SBI) per topic (n = 26,996 comments, 9 subreddits). Topics sorted ascending by SBI (least balanced at top). The dotted vertical line at x = 0.5 marks perfect balance. Bar colour transitions from red (low SBI, one-sided) to yellow (moderate) to green (balanced).

2b. MSDG — Minority Stance Defensiveness Gap

MSDG measures whether minority-stance users write more defensively than majority-stance users. It is computed as mean defensiveness of minority-stance comments minus mean defensiveness of majority-stance comments, where defensiveness captures language that anticipates social punishment or backlash (e.g., disclaimers, hedging, self-distancing). A positive value suggests minority users moderate their expression in response to the community's dominant stance. The n values shown indicate the number of minority-stance comments available per subreddit.

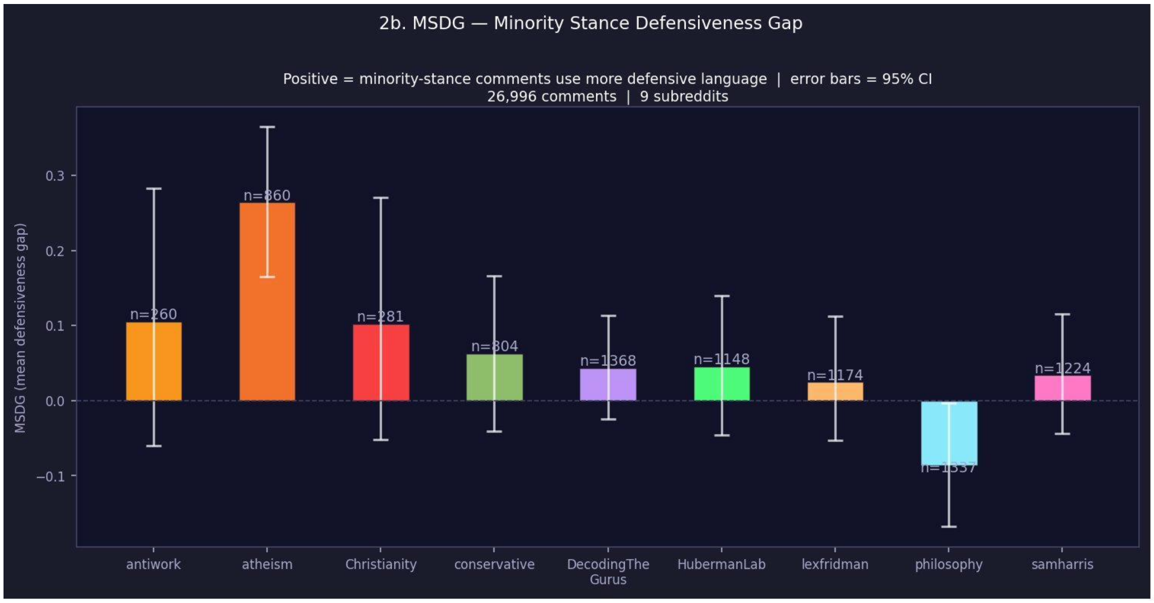

Figure 2b. Minority Stance Defensiveness Gap (MSDG) per subreddit (n = 26,996 comments, 9 subreddits). Positive bars: minority-stance comments use more defensive language than majority-stance. Negative bars: majority-stance comments are more defensive. Error bars show 95% CI. n values show minority comment counts per subreddit.

3a. RDB — Reply Direction Bias

RDB measures the fraction of replies each user group directs at same-stance comments. Three bars are shown per subreddit: Majority-stance users (green), Minority-stance users (red), and an Aggregate bias bar (blue) showing the overall tendency. A value of 0.5 would indicate random, stance-blind reply behaviour. Values above 0.5 would indicate users preferentially reply to same-stance commenters (echo-chamber behaviour). Values below 0.5 indicate users preferentially engage across stances.

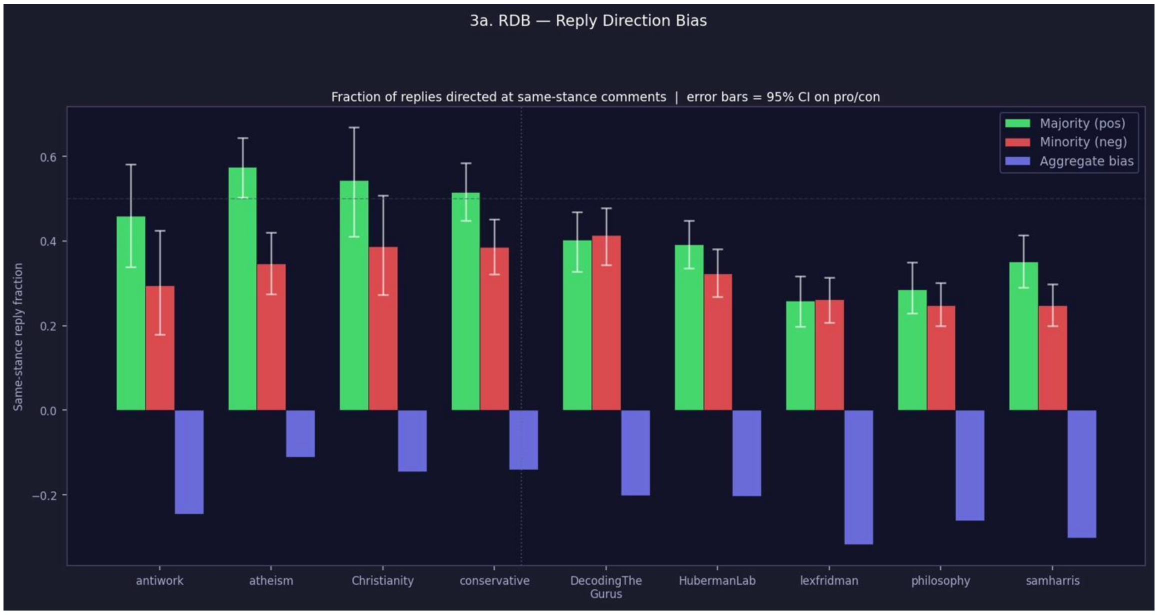

Figure 3a. Reply Direction Bias (RDB) per subreddit. Y-axis: fraction of replies directed at same-stance comments. Green = Majority-stance users, Red = Minority-stance users, Blue = Aggregate bias. Dotted line at 0.5 marks random chance. Vertical dotted divider separates the two community groups.

3b. uRDB — User-Level Reply Direction Bias

uRDB applies the same logic as RDB but averaged at the individual-user level: each user who made at least four replies in a thread contributes one data point equal to their personal same-stance reply fraction, before these are averaged across all qualifying users in the subreddit. This prevents high-volume users from dominating the aggregate signal.

Figure 3b. User-Level Reply Direction Bias (uRDB) per subreddit. Y-axis: mean per-user same-stance reply fraction. Dotted line at 0.5 marks random chance (shown at top of the plot). Error bars show 95% CI.

4a. EAS — Emotional Amplification Score

EAS measures whether more emotional comments receive more upvotes, using Spearman rank correlation (ρ) between emotion scores and upvote counts. Three emotions are tested per subreddit: anger (red), anxiety (yellow-green), and disgust (also red/salmon). A positive ρ means the community upvotes emotionally charged content. A negative ρ means emotional comments are downvoted relative to calmer ones. An asterisk (*) marks correlations that survive Benjamini–Hochberg FDR correction ($p < 0.05$).

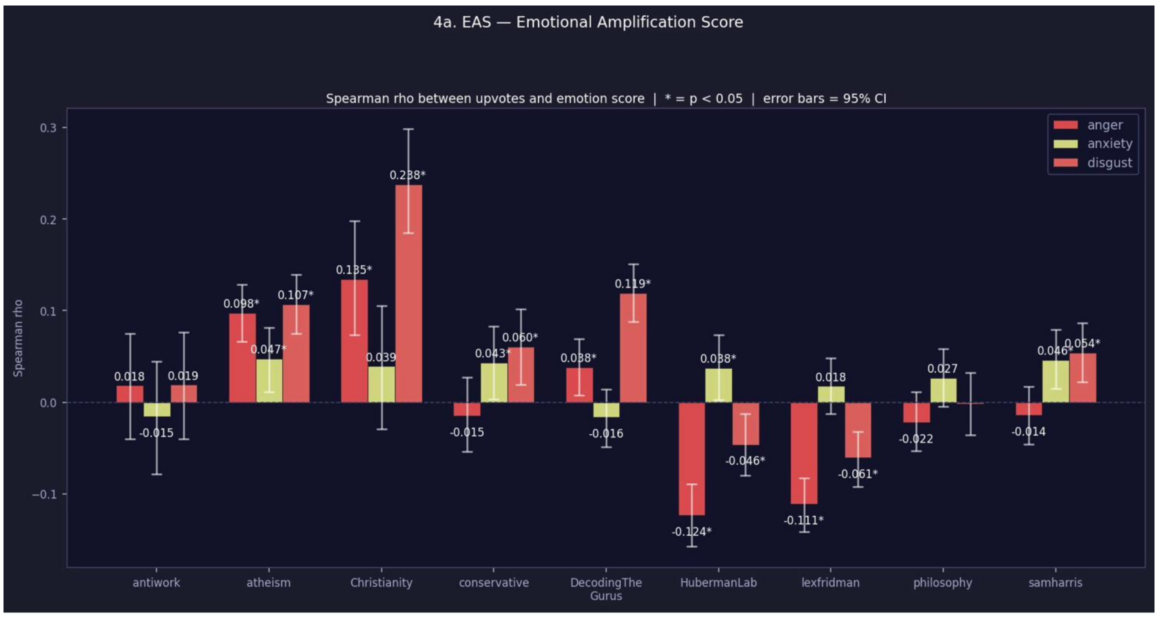

Figure 4a. Emotional Amplification Score (EAS) per subreddit. Y-axis: Spearman ρ between upvotes and emotion score. Red = anger, yellow-green = anxiety, salmon/dark-red = disgust. Asterisk (*) = $p < 0.05$ after FDR correction. Grey dashed line at ρ = 0.

4b. CSAD — Cross-Stance Anger Differential

CSAD compares mean anger scores across three types of reply interactions within each subreddit: Majority-stance users replying to Minority-stance comments (Maj→Min, red), Minority-stance users replying to Majority-stance comments (Min→Maj, blue), and users replying to comments of the same stance as themselves (Same-stance baseline, yellow-lime). If cross-stance interactions (red, blue) are angrier than the same-stance baseline (yellow), the community shows heightened hostility toward opposing viewpoints.

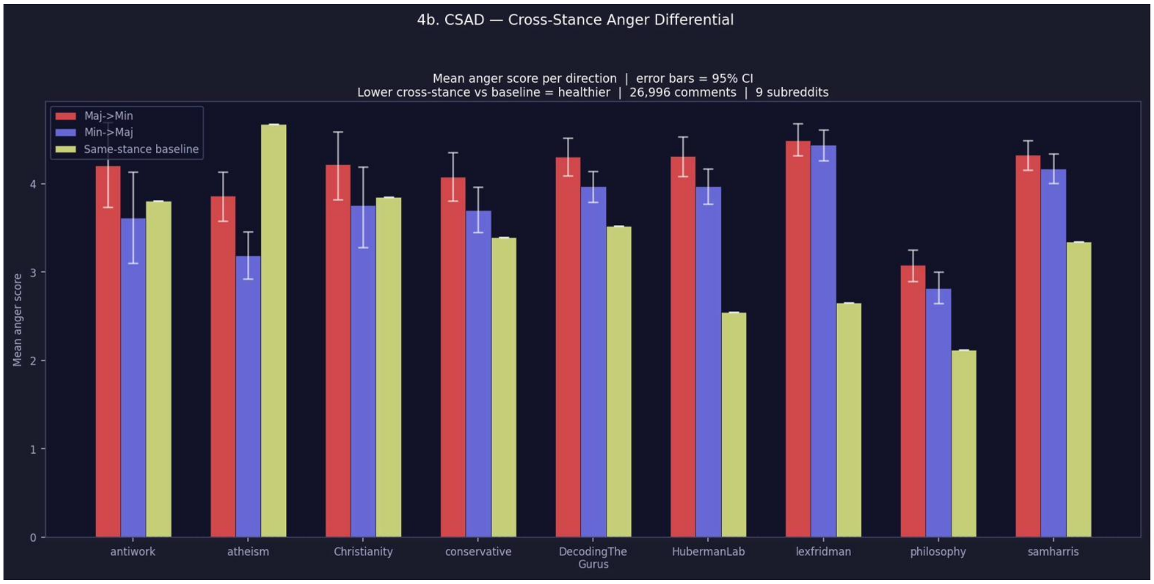

Figure 4b. Cross-Stance Anger Differential (CSAD) per subreddit. Mean anger score (0–10) per reply direction: red = Maj→Min, blue = Min→Maj, yellow-lime = same-stance baseline. Error bars show 95% CI.

5. TD — Toxicity Differential

TD follows the same directional structure as CSAD but uses toxicity scores instead of anger. Toxicity captures hostile, aggressive, or abusive language directed at the parent comment or its author (insults, slurs, dehumanisation, threats). The same three bars are shown: Maj→Min (red), Min→Maj (blue), and same-stance baseline (yellow-lime).

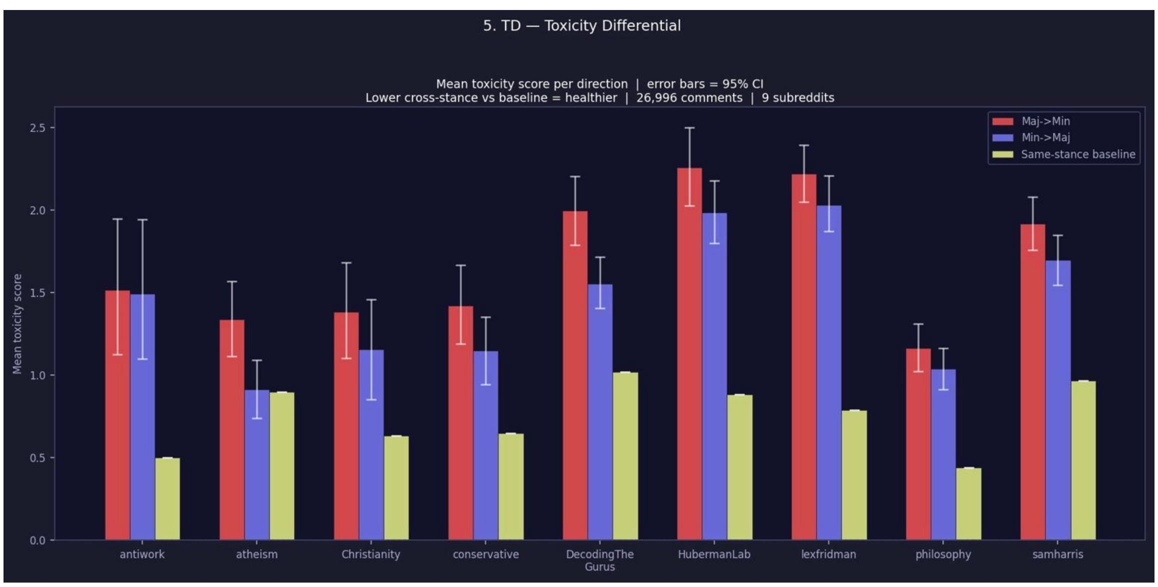

Figure 5. Toxicity Differential (TD) per subreddit. Mean toxicity score (0–10) per reply direction: red = Maj→Min, blue = Min→Maj, yellow-lime = same-stance baseline. Error bars show 95% CI.

Additional Supplementary Material

S2 Post Title Analysis Prompt

The following system prompt is used in Stage 2 (post title analysis). It classifies each post title into the fixed topic taxonomy and assigns a discussion score. The user message consists only of: Post title: "<title>".

System Prompt:

You are a research assistant analyzing Reddit post titles for discourse research.

You will be given only a post title. Return ONLY valid JSON (no markdown, no prose).

OUTPUT SCHEMA:

```
{
  "topic": { "top": "<TOP>", "mid": "<MID>", "leaf": "<LEAF or null>" },
  "secondary_topics": [{"top": "...", "mid": "...", "leaf": "..."}],
  "topic_confidence": <float 0.0-1.0>,
  "secondary_confidence": [<float>],
  "discussion_score": <float 0.0-1.0>,
  "discussion_reason": "<one short sentence>"
}
```

TASK A - TOPIC CLASSIFICATION (STRICT, PREDEFINED TAXONOMY)

Use ONLY labels from the fixed taxonomy in constants.py.

Allowed top labels: politics, economics, religion, technology, science, philosophy, society, UNCLEAR.

If ambiguous: topic = {top: UNCLEAR, mid: null, leaf: null}, topic_confidence <= 0.30.

TASK B - DISCUSSION SCORE

0.00-0.20: Non-debate content | 0.21-0.40: Mostly informational

0.41-0.60: Some controversy | 0.61-0.80: Clear opposing viewpoints likely

0.81-1.00: Highly polarizing conflict likely

Reserve scores >0.80 for strongly polarizing moral/identity/political conflict.

S3 Comment Analysis Prompt

The following prompt is used in Stage 6 (comment analysis). The static instruction block (everything above GROUP CONTEXT) is cached as the Anthropic system prompt for ~90% token cost reduction across repeated calls in the same run. The dynamic template is sent as the user message per comment, with GROUP_CONTEXT, PARENT_TEXT, PARENT_TOPIC, and TEXT filled in for each call.

System Prompt (static, cacheable portion):

You will be given a single Reddit post with full prior thread context.

Use all prior context for interpretation; score and analyze only the final comment.

Scale: 0-10 unless otherwise stated.

0 = completely absent | 1-3 = weak | 4-6 = moderate | 7-9 = strong | 10 = extreme

ATTRIBUTES:

comment_type (array): question, information, opinion, sarcasm, meta-commentary,

off-topic, advice, correction, anecdote, low effort, humour, other.

topic -> label: Reuse parent label if same topic; else create new hierarchical label.

topic -> stance -> value (-10 to +10 or N/A):

-10 to -8 = strong explicit disagreement with emotional language

-7 to -4 = clear respectful disagreement

-3 to -1 = mild implicit disagreement

0 = neutral / tangential

1 to 3 = mild implicit agreement

4 to 7 = clear direct agreement

8 to 10 = strong explicit agreement with emotional language

N/A = ambiguous or insufficient context

epistemic_risk -> claim_strength (0-10 | N/A for opinion/question)

epistemic_risk -> evidence_quality (0-10 | N/A for opinion/question) epistemic_risk -> reasoning_depth (0-10 | N/A: does NOT require empirical evidence)

toxicity   (0-10): Hostile language directed at the parent comment/author.

discrediting(0-10): Undermining the parent's legitimacy/motives without engaging the argument.

defensiveness(0-10): Anticipatory softening (e.g. 'I know I'll get downvoted but...').

civility   (0-10): 0 = extremely rude, 5 = neutral, 10 = exceedingly polite.

emotion -> anger, anxiety, disgust (0-10): Emotional tone in the comment overall.

GROUP CONTEXT: {{GROUP_CONTEXT}}

PRIOR CONTEXT: {{PARENT_TEXT}}

COMMENT TO ANALYZE: {{TEXT}}

S4 V3 Metric Constants

Key constants from constants.py governing V3 metric computation.

Constant	Value	Description
V3_STANCE_THRESHOLD	2	Comments with |stance| <= 2 excluded from stance metrics
V3_MAJORITY_MIN_FRACTION	0.60	Majority must hold >= 60% of classified comments
V3_MSDG_MIN_MINORITY_PER_TOPIC	10	Min minority comments per topic for MSDG
V3_MSDG_MAX_SBI	0.4	Topics with SBI above this excluded from MSDG
V3_URDB_MIN_REPLIES_PER_USER	4	Min replies a user must have to count in uRDB

V3_EAS_TOPIC_MIN_COMMENTS	50	Min comments per topic for topic-level EAS
V3_EAS_BOOTSTRAP_ITERS	1000	Bootstrap iterations for EAS confidence intervals
COMMENT_ANALYSIS_MODEL	claude-haiku-4-5-20251001	LLM model for comment annotation
COMMENT_PREPROCESSING_MAX_SAMPLE	100	Max comments sampled per post
COMMENT_PREPROCESSING_MIN_CONTENT_LENGTH	20 chars	Min comment length to include

S5 Data Collection Runs

The combined dataset of 26,996 comments drawn from 9 subreddits was produced by two separate scraping runs with different community targets and preprocessing parameters.

scrape_006 — Political, Social, and Religious Communities

Ten subreddits were targeted: conservative, lateStageCapitalism, antiwork, decodingthegurus, trump, Christianity, atheism, technology, philosophy, and sociology. Preprocessing used strict filters (min_discussion_score = 0.8, min_topic_peers = 6, top_n_per_subreddit = 30), which caused significant attrition for smaller communities. 156 posts were selected with highly uneven distribution — atheism and conservative reached the cap of 30 while sociology produced only 2.

The comment analysis ran in two phases and was aborted when the Anthropic API credit balance was exhausted at approximately 45% completion. Only 4 of the 10 subreddits had their comments analysed: atheism (3,382 comments, 30 posts), conservative (2,307, 24 posts), antiwork (1,069, 8 posts), and Christianity (916, 8 posts) — totalling 7,674 successful analyses across 70 posts. The remaining 6 subreddits (decodingthegurus, lateStageCapitalism, philosophy, sociology, technology, trump) were not reached before the credit ran out.

scrape_007 — Intellectual Podcast Communities

Five subreddits were targeted: samharris, decodingthegurus, lexfridman, HubermanLab, and philosophy. Preprocessing used relaxed filters (min_discussion_score = 0.6, min_topic_peers = 1), which ensured all five communities reached the 30-post cap (150 posts total). A single comment analysis run processed all 18,864 filtered comments without interruption (approximately 2 hours 9 minutes, effectively zero errors), yielding 18,853 successful analyses. Per-subreddit counts were: lexfridman 4,333; decodingthegurus 3,802; samharris 3,718; philosophy 3,624; HubermanLab 3,376.

Combining the Two Runs

The combined visualisations load pre-computed per-subreddit metrics from both runs and place all 9 subreddits on the same axes. The two scrapes' subreddit sets are completely disjoint (despite decodingthegurus and philosophy appearing in both scraping runs, only scrape_007's comment analyses were available for both), so the dictionaries are merged with a simple union with no key collisions.

Metrics were not recomputed on the combined dataset — each subreddit's metrics reflect its own scrape's context. Sample sizes differ substantially between the two groups: scrape_006 subreddits average ~1,919 comments each while scrape_007 subreddits average ~3,771. Additionally, scrape_006 used stricter post-selection filters, meaning the communities in that run contributed fewer posts and had less balanced topic coverage. These asymmetries should be considered when comparing metrics across the two groups.

Suplementary Material

S1 Comment LLM Prompt

The following is the LLM prompt I will use on the comment.

You will be given a single Reddit with parent comment or post and relevant metadata.

Your task is to extract structured attributes used to measure discourse dynamics

(e.g. selective engagement, discreditation, self-censorship, emotional amplification).

Only use information explicitly present in the text.

Do not infer beliefs, intent, or factual accuracy beyond what is stated.

Do not use external knowledge.

If an attribute is not applicable or cannot be inferred, return "N/A".

Return valid JSON only.

DEFINITIONS:

Topic label: Given a parent topic label, decide whether the comment addresses the same substantive topic or introduces a new, meaningfully different topic. If it is the same topic, reuse the parent topic label verbatim. If it introduces a new topic, create a high-level topic label that can reasonably encompass future replies. Topic labels should be concise and general (2–6 words).

Topic stance: The author’s expressed position toward the identified topic.

- −1 = strongly opposed
- 0 = neutral / unclear
- +1 = strongly supportive

Topic confidence: the confidence of which you applied the topic stance rating.

Discrediting:

Language that undermines the legitimacy, intelligence, motives, or moral standing of an opposing view instead of substantively engaging with its argument.

Defensive:

Language that anticipates social punishment or conflict and strategically softens or shields expression (e.g. disclaimers, fear of backlash, self-distancing).

Substantiveness:

The degree to which the text contributes meaningful reasoning, evidence, or explanation.

Toxicity:

Hostile or aggressive language such as insults, slurs, dehumanization, or threats.

Politeness:

Surface-level civility and respectful phrasing, independent of stance.

Emotion intensities:

Strength of expressed emotional tone (anger, fear, outrage, anxiety, disgust)